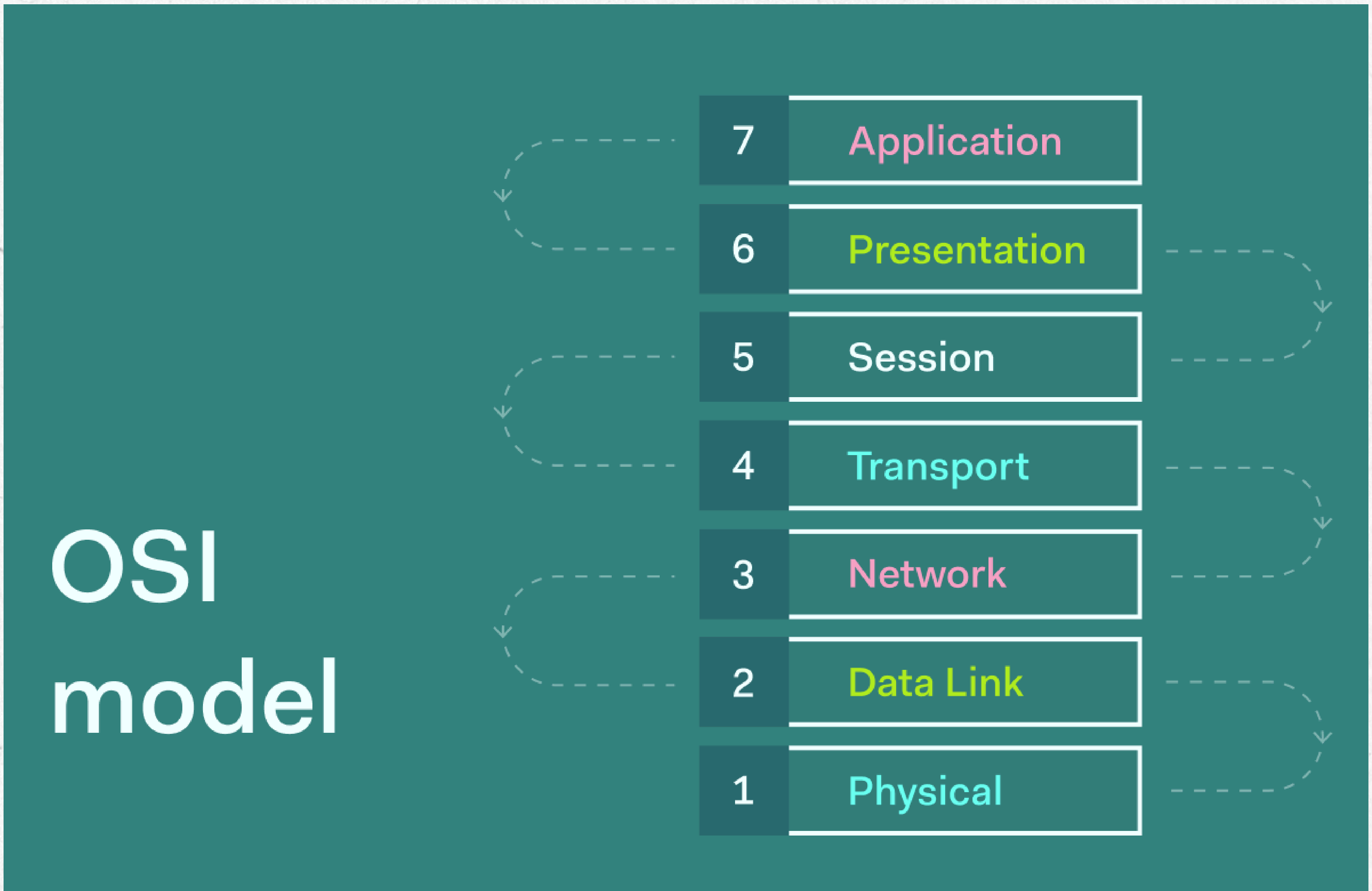
OSI Layers and it's common attacks

Introduction

The OSI Model helps us understand how data moves across networks.

But each layer also introduces unique security vulnerabilities.



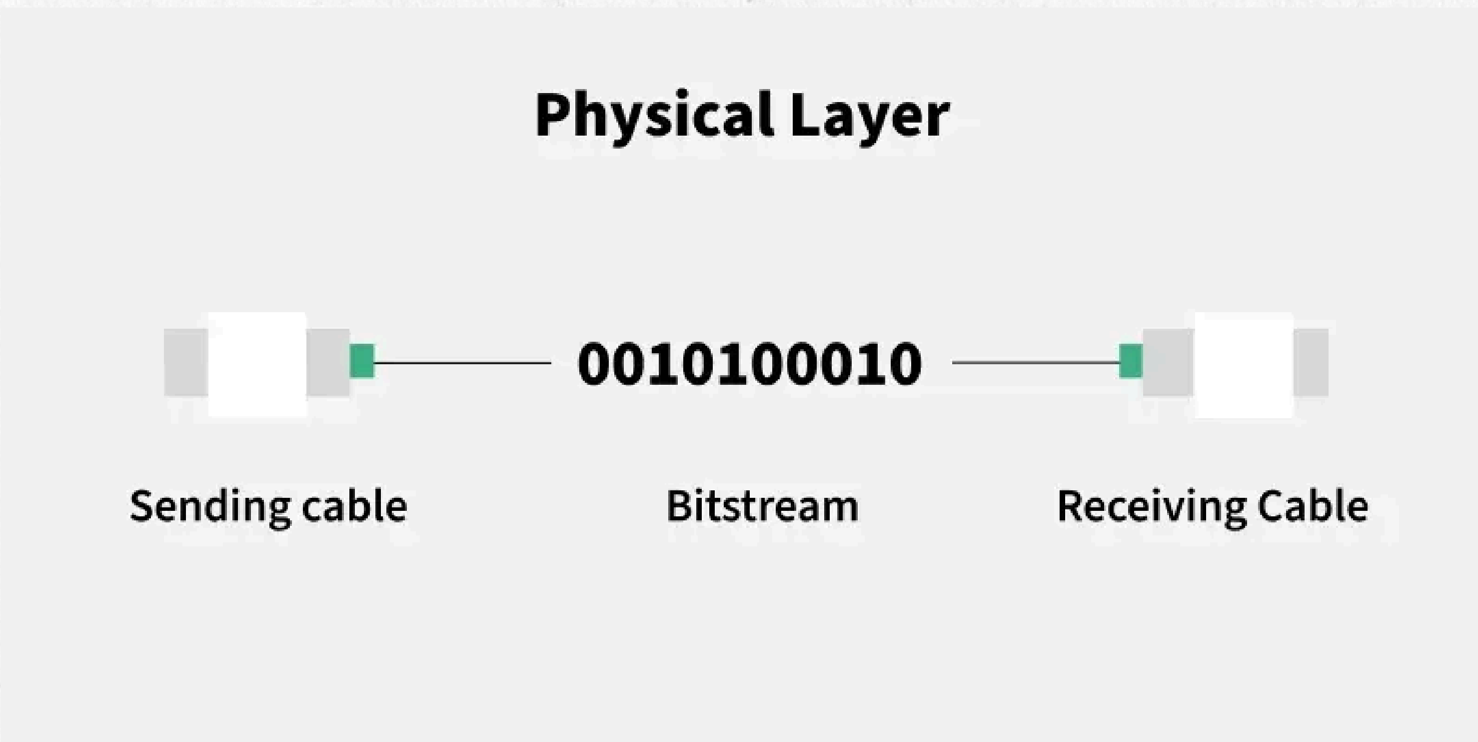
1. Physical Layer

Function:

Transfers raw bits (0s and 1s) through physical hardware — cables, switches, Wi-Fi signals.

Common attacks:

- Cable tapping – accessing data by tapping network cables.
- Signal jamming – blocking Wi-Fi or wireless signals.
- Device theft – stealing/disconnecting hardware.
- Hardware tampering – inserting malicious devices.



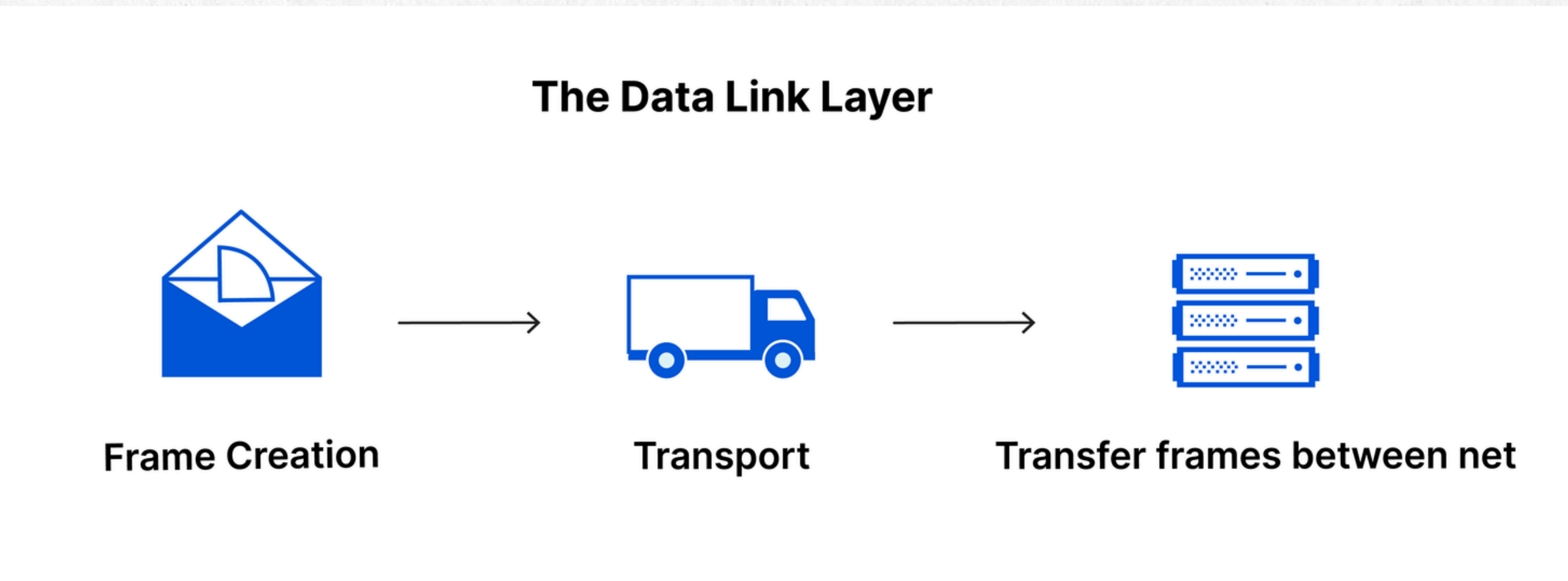
2. Data Link Layer

Function:

Handles communication within a local network using MAC addresses and frames.

Common attacks:

- ARP Spoofing – attacker tricks devices to redirect traffic.
- MAC Spoofing – pretending to use another device's MAC.
- VLAN Hopping – bypassing VLAN boundaries.
- Switch flooding – forcing the switch into broadcast mode to sniff traffic.



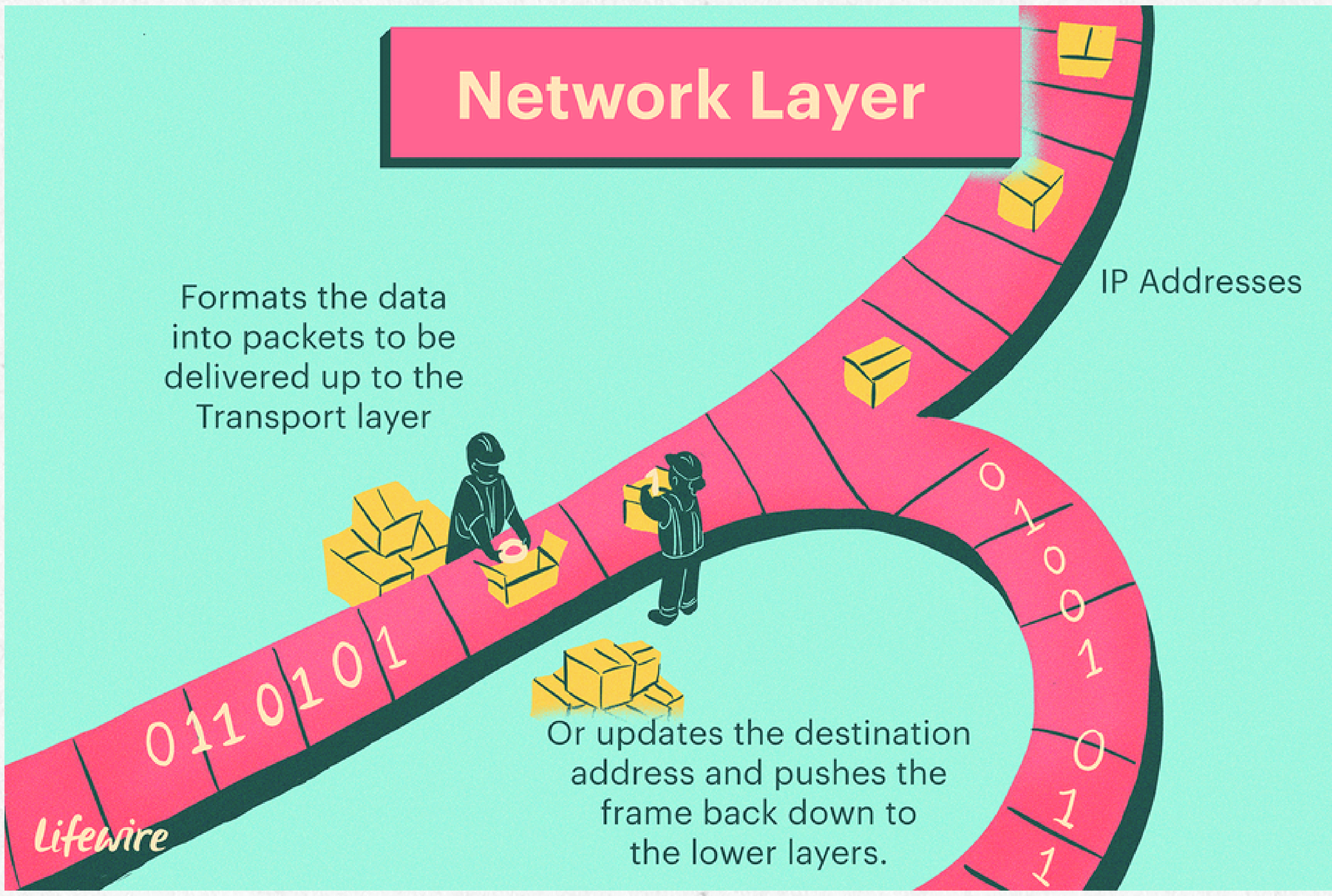
3. Network Layer

Function:

Routes data between networks using IP addresses

Common attacks:

- IP Spoofing – faking IP addresses.
- DDoS – overwhelming a network with traffic.
- Route Hijacking – manipulating routing paths.
- Packet fragmentation attacks – bypassing filters using split packets.



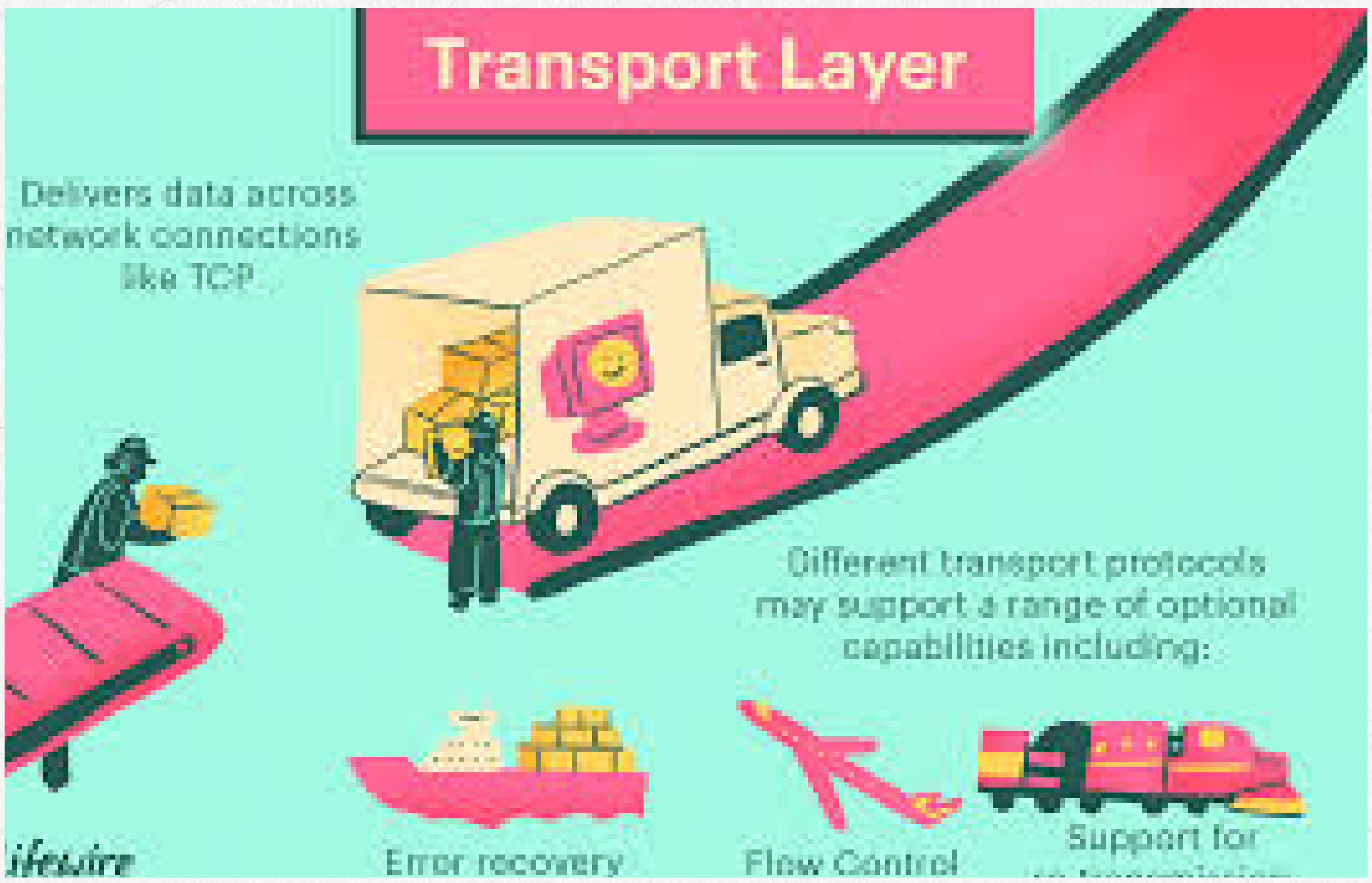
4. Transport Layer

Function:

Manages end-to-end communication using TCP/UDP and port numbers.

Common attacks:

- SYN Flood – flooding a server with fake TCP requests.
- UDP Flood – overwhelming with random UDP packets.
- Session Hijacking – taking over active connections.
- Port Scanning – finding open ports to target.



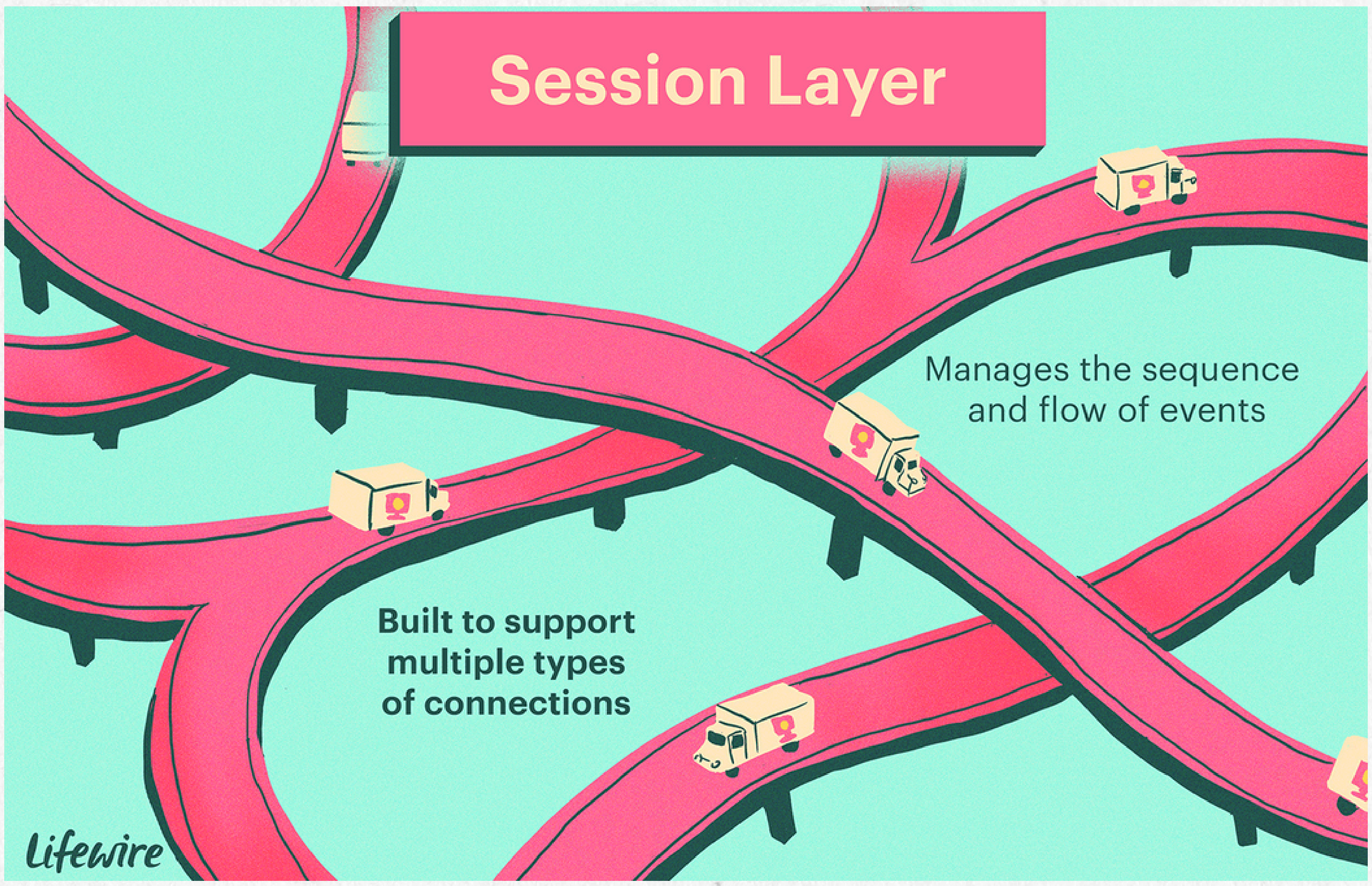
5. Session Layer

Function:

Creates and manages sessions (logins, connections).

Common attacks:

- Session Hijacking – stealing active sessions.
- Session Replay – reusing captured session tokens.
- Session Fixation – forcing user to use attacker-controlled session ID.



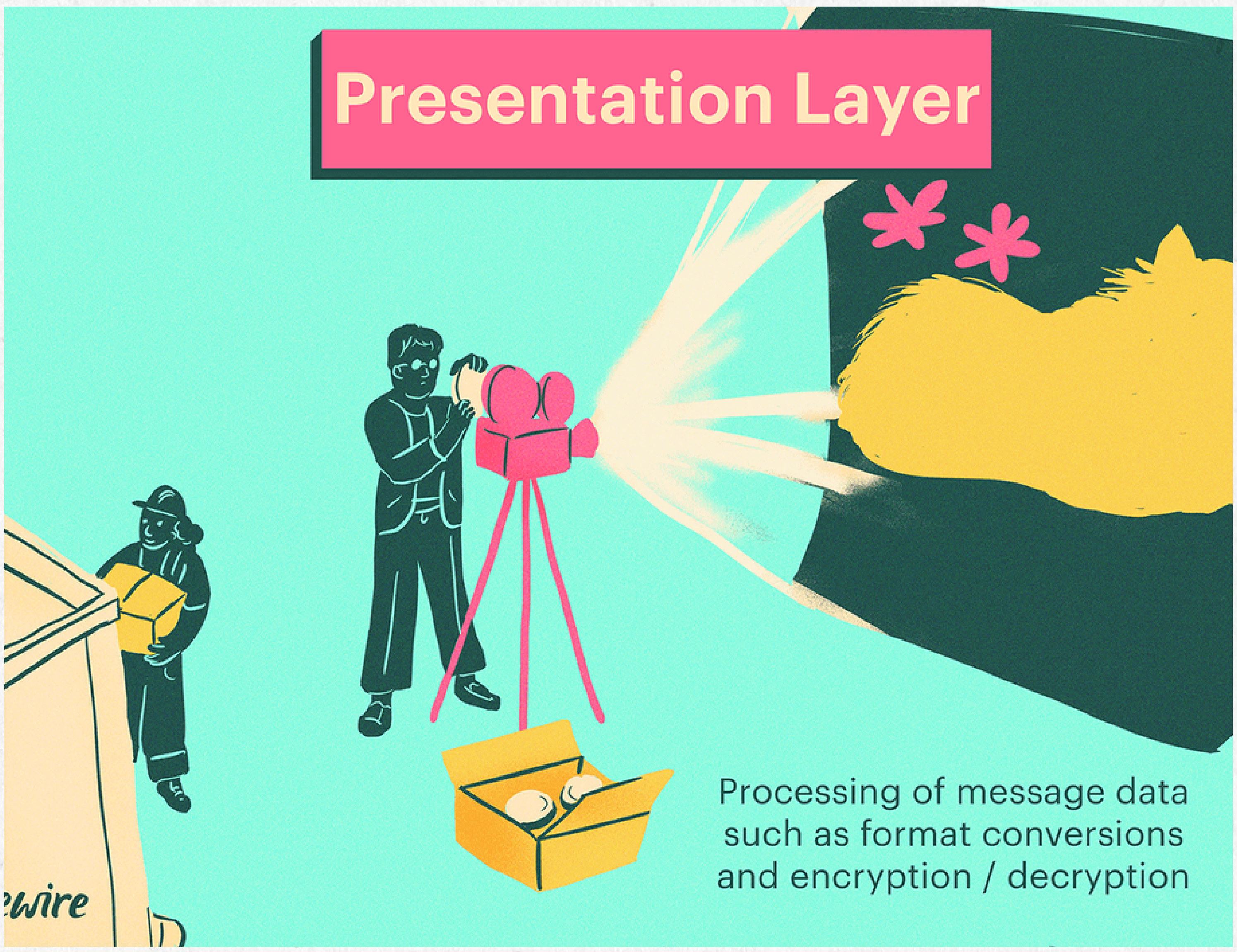
6. Presentation Layer

Function:

Formats data (encryption, compression, encoding)..

Common attacks:

- SSL Stripping – downgrading HTTPS to HTTP.
- Certificate attacks – using fake certificates.
- Encoding/decoding exploits – manipulating how data is parsed.
- Serialization attacks – exploiting unsafe data formats.



7. Application Layer

Function:

Provides services users interact with — HTTP, DNS, email, APIs.

Common attacks:

- SQL Injection – injecting malicious database queries.
- XSS (Cross-Site Scripting) – injecting harmful scripts.
- CSRF – forcing users to perform unwanted actions.
- API Attacks – abusing weak API endpoints.
- Phishing / Malware – tricking users into giving access.